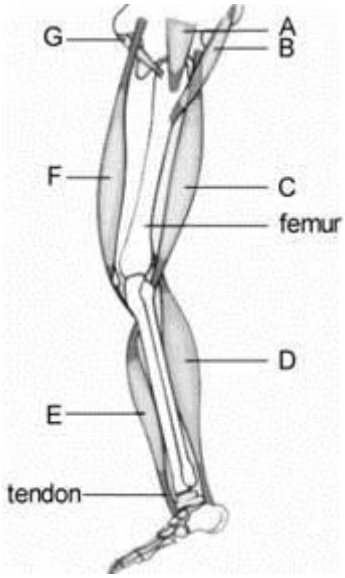
Q1. (a) Diagram 1 is a simplified drawing of bones and muscles in a left leg viewed from the outer side.



(i) Muscle A contracts. What effect does this have on the leg?

.....

.....

1 mark

(ii) Muscles are found in antagonistic pairs. Write the letters of the pair of muscles which control the bending and straightening of the leg at the knee.

.....

1 mark

(iii) When one muscle of an antagonistic pair contracts the other muscle does not relax completely, but maintains some tension.

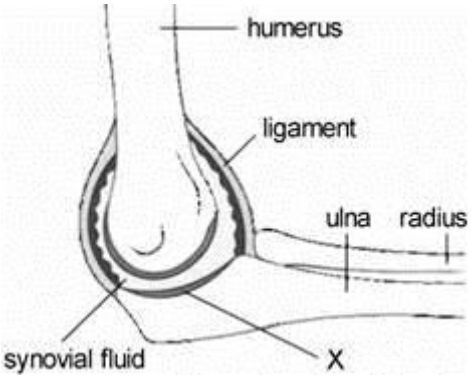
What is the advantage of maintaining tension in both muscles.

.....

.....

1 mark

(b) Diagram 2 shows the elbow joint. Tissue X covers the ends of the bones at the joint.



(i) Give the name of tissue X. .....

1 mark

(ii) Osteoarthritis is a very painful condition. In the joints of people with osteoarthritis small pieces of tissue X break off.
Suggest **two** effects this could have on the joint.

.....

.....

.....

2 marks

(c) Tendons are shown in diagram 1 and ligaments are shown in diagram 2. Ligaments can stretch much more than tendons.

(i) Explain why ligaments at the elbow need to stretch.

.....

.....

1 mark

(ii) Explain why it is necessary that tendons hardly stretch at all when a muscle contracts.

.....

.....

1 mark
Maximum 8 marks

Mark schemes

Q1.

(a) (i) it pulls the leg **or** the femur outwards **or** to the side

accept 'it lifts the leg outwards or to the side'
or 'the leg swings to the side or outwards'

*do **not** accept 'it lifts the leg'*
or 'it pulls the leg upwards'

1

(ii) C and F

letters may be in either order

***both** letters are required for the mark*

1

(iii) any **one** from

- it allows controlled **or** slow movements

  accept 'so that movements are not jerky'
  ***or** 'the relaxed muscle provides resistance to movement'*

- it allows small movements

  accept 'the partially relaxed muscle holds the limb in position'

  accept 'if they relaxed completely the person would fall over'
  ***or** 'it allows more rapid responses'*

1

(b) (i) cartilage

1

(ii) any **two** from

- ends of bones will rub against each other **or** become worn

  accept 'there will be friction at the joint'
  *do **not** accept 'the bones will hit each other'*

- less smooth movement in the joint **or** stiffness in the joint

  accept 'harder to move'

- inflammation **or** swelling

  *do **not** accept 'pain'*

2

(c) (i) to allow the arm to bend **or** move at the joint

accept 'to allow the arm to move'
*accept 'so they do **not** snap when the arm bends'*
***or** 'they need to be different lengths when the joint moves'*

*do **not** accept 'to allow bones to bend'*

1

(ii) if they stretched they would not pull as much on the bones

accept 'they would have less effect on bones'
***or** 'controlled movement would not be possible'*
***or** 'they would not pull as much'*

*do **not** accept 'they would not pull on the bone'*

or *'they would not hold bones'*

1

[8]